

Deepak Mallampati

(330) 554-0770 | deepakmallampati00@gmail.com | linkedin.com/in/deepak-mallampati | Kent, OH

PROFESSIONAL SUMMARY

Software engineer (MS CS, 2.5+ years) specializing in enterprise AI deployment, applied ML pipelines, and customer-facing technical delivery. Production experience in Python, RAG systems, agentic workflows, FastAPI, and Docker. Strong cross-functional communication for both technical teams and business stakeholders. Background includes database engineering (SQL/T-SQL optimization, CI/CD automation) and enterprise data pipeline work - applicable to Resilinc's supply chain data integration requirements.

CORE COMPETENCIES

Enterprise AI Deployment • Python / FastAPI / Docker • RAG Systems • Agentic Workflow Development • SQL / T-SQL / Database Engineering • Data Pipeline Design • REST API Integration • CI/CD (Jenkins) • Cross-Functional Stakeholder Delivery

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer - Healthcare Technology, AI Consulting

- Designed and delivered **enterprise AI systems** for healthcare workflows; RAG systems (~35% precision gain), agentic workflows (~25% stability improvement), predictive ML (15-20% recall improvement).
- Built data pipelines for EHR extracts and support ticket logs; raised dataset reliability above **98%**.
- Packaged AI as **FastAPI/Flask RESTful services** with Docker; REST API integration with enterprise systems; reduced post-deploy defects ~**30%**.
- Translated ambiguous business requirements into technical architecture and implemented production solutions.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern)

- Optimized SQL and **T-SQL stored procedures** powering Order-to-Cash platform; reduced query execution time ~**20%**, retrieval latency ~**25%**.
- Built **Jenkins CI/CD pipelines** for schema deployments; reduced deployment errors >**30%**.
- Designed **Grafana dashboards** using DMVs for long-running queries and deadlock tracking.
- RBAC implementation and audit logging for compliance-sensitive financial modules.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer

- Transformer-based summarization; RAG document Q&A. Flask API for production non-technical users.

KEY PROJECTS

Agentic Claims Processing & Fraud Risk Intelligence

Multi-agent pipeline delivering complex enterprise AI use cases: Intake → Validation → Consistency → Duplicate Detection → Risk Scoring. Schema-validated agent contracts, RAG-grounded validation, explainable reasoning traces. Analog to Resilinc's agentic disruption intelligence and supplier risk assessment workflows.

Geospatial Road Network Analysis - Enterprise Data Pipeline

Parsed raw OpenStreetMap XML into weighted NetworkX DiGraph; Haversine edge weighting; Dijkstra routing; betweenness centrality on 20k-node subgraph; Louvain community detection. ~30% runtime reduction via subgraph scoping. Demonstrates complex data pipeline engineering at enterprise scale.

EDUCATION

Master of Science, Computer Science - Kent State University, Ohio, USA

Aug 2023 - May 2025

SKILLS

Languages: Python, SQL (T-SQL, PostgreSQL, SQLite), TypeScript, JavaScript

AI & ML: RAG, agentic workflows, LangChain, LlamaIndex, scikit-learn, XGBoost, PyTorch, HuggingFace

Data: Pandas, NumPy, NetworkX (graph analytics); familiar with data pipeline design

Infra: Docker, FastAPI, Flask, Jenkins, Grafana, REST APIs

Note: Currently US-based (F-1 OPT); role requires India residency - please verify logistics